

ROBERT BRUNNER

Full Stack Software Developer

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🌐 <https://www.robertbrunner.dev/>

🔗 <https://github.com/robert-brunner>

in <https://www.linkedin.com/in/robert-e-brunner/>

Enthusiastic developer with experience spanning embedded communications, mobile application development, hardware diagnostics, 3D modeling, multimedia production, and visual design.

SKILLS

- | | | | |
|------------------------------|--------------------|-------------------------------|---------------------------|
| • React.js | • Node.js | • Fusion 360 | • MPLAB X |
| • Ionic/Capacitor | • C# | • Adobe After Effects | • Embedded Systems |
| • Android Development | • .Net Core | • Adobe Premier | • PIC1637/505 ASM |
| • C++ | • SQL | • Figma (UI/UX Design) | • UART/ESP32/RF |

EXPERIENCE

Embedded Systems & Software Developer

Applied Wireless

**Full Time
August 2024 - Present**

- Built a cross-platform mobile app from scratch (Ionic/React + Capacitor) with BLE, real-time packet handling, and GPS-triggered auto-lock, collaborating closely with the lead hardware engineer on the embedded firmware side.
- Engineered end-to-end comms: JS → BLE → C++ (ESP32) → UART → PIC (ASM) using 10-byte packets, with checksums, state sync, and exponential backoff.
- Designed and implemented professional UI/UX in Figma (dark mode, dynamic formatting, real-time brake force visualization) and created UI animations/loading screens in After Effects, implemented with Lottie JSON.
- Managed Android deployment and APK signing via Capacitor and Gradle.
- Self-taught Fusion 360 on the job to produce accurate .STEP file models and enclosure prototypes; utilized CNCs and soldered boards/microchips from schematics.
- Validated hardware with oscilloscopes, analyzers, and signal generators; self-taught hex/binary protocol design and embedded debugging while delivering production-ready features.

Full Stack Software Developer

NewForce

**Internship
June 2022 - January 2023**

Intensive full-time 6-month software development immersive training program focusing on full stack (C#/.NET) development fundamentals and problem solving. The final half of the program executed in a simulated company environment with Scrum methodology.

- Applied JavaScript, HTML, and CSS fundamentals to build a feature-rich social media dashboard
- Created RESTful Web API with C#/.NET Core and connected it to a React front-end
- Developed a blog management platform in ASP.NET, MVC, and Razor templates in Visual Studio 2022
- Built and interacted with databases using SQL and ADO.NET
- Designed applications through white boarding dependencies and building ERD's
- Leveraged native ES6 module bundling to build DRY, reusable components
- Designed and built a web app with React using hooks capable of manipulating data using the C.R.U.D. methods. In addition, I incorporated aspects of CSS and Sass to make it more visually appealing
- Applied object-oriented programming fundamentals through team-based projects that reflect real world business problems
- Collaborated remotely on projects using Slack and Zoom
- Managed source code version control with Git/ GitHub

Lead Alarm Technician/Solar Sight Surveyor

NorthStar Alarm Services, LLC

**Full Time
May 2014 - August 2024**

- Trained technicians and customers on technical systems, troubleshooting, and app-based controls; delivered exceptional problem-solving support.
- Managed installations, programming, and customization of smart home/security devices, demonstrating strong technical aptitude and customer focus.

PROJECTS

Real-Time Embedded Control Platform

Architected and solo-developed a real-time Android application (Ionic React + Capacitor) for an embedded vehicle control system. Designed the complete UI/UX in Figma, including dark mode, animated Lottie assets, dynamic status visualization, integrated help systems, and real-time telemetry displays. Designed a custom 10-byte bidirectional communication protocol and developed the mobile-to-hardware communication stack (JavaScript → BLE → ESP32 C++ → UART), including checksum validation, synchronization, retry/backoff logic, and reliable command delivery between the Android application and embedded controller.

Cute-N-Furry - (Tinder for dogs)

Full-stack React + C#/.NET app with authentication, profile editing, swipe-based matching, and Cloudinary image uploads.

🔗 <https://github.com/robert-brunner/FFBNs>

MidEastern Whetherfleura

Complete CRUD React application built in two weeks using Sass and json-server for persistent storage of creature sightings.

🔗 <https://github.com/robert-brunner/MidEasternWhetherfleura>

EDUCATION

NewForce

Software Development Intern

CERTIFICATIONS

Front-End Development - 8/26/22

Full-Stack Development - 12/16/22

Server-Side Development - 12/16/22

MountWest Community and Technical College

COMMUNITY SERVICE

Montpelier City Volunteer Fire Dept.

Bear Lake County Volunteer Fire Dept.

Volunteer Firefighter 2012-2014

NOTABLE ACHIEVMENTS

Eagle Scout